Sarah Barnes Haddix

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Education

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA
B.S. Computer Science, B.S. Mathematics — 08/2022 - 05/2026
Graduated with Honors in Computer Science (completed and defended a senior honors thesis)
Graduated with Distinction (awarded for a cumulative GPA of 3.50 or higher)
Member, Honors Carolina (academic honors program admitting under 10% of each incoming class)
Dean's List: Fall 2022 – Spring 2026 (8 semesters) Major GPA: 3.90/4.00 Overall GPA: 3.79/4.00

Professional Experience

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA
Undergraduate Research Assistant — 04/2024 - 06/2026

- Worked with Computer Science Department Chair Dr. James H. Anderson's Real-Time Systems Research Group, focusing on real-time operating systems, AI, and GPU scheduling
- Assisted graduate students in conducting experiments, for example exploring GPU memory constraints and overload behaviors, and provided feedback and proofreading on research papers
- Co-authored a real-time robotics paper accepted at ICRA25; major contributions included the development and modification of software in Python and C++ to control a simulated drone leveraging Linux and ROS and the optimization of a path planning neural network using Tensorflow and ONNX

Owl Cyber Defense — Morrisville, North Carolina, USA
Engineering Intern — 06/2025 - 06/2026

- Worked in Rust and Java to develop cross-domain network security solutions for US government cybersecurity applications
- Added secure chat capabilities to existing products including the V2CDS, an NCDSMO Baseline Certified technology, using open source projects like Apache Nifi and OpenFire to handle streaming XML/XMPP data. Work on this project also included building a PostgreSQL-backed dashboard and management of Linux environments and virtual machines
- Held a U.S. Secret clearance

Axai Tech — Cape Town, Western Cape, South Africa
Project Intern — 06/2024

- Developed and applied deep learning models for protein sequence analysis and generation, supporting research into minimally invasive cancer therapies.
- Developed and trained multiple deep learning models, including LSTM-based architectures and transformer models, using PyTorch and Hugging Face. Performed data preprocessing, cleaning, transformation, and exploratory analysis using Python, NumPy, and Pandas to prepare biological datasets for machine learning applications

Scale AI — Remote
Software Engineer for AI Training Data — 05/2024 - Present

- Contributing to coding-related AI training data in mainly Java, Python, and Swift used to finetune modern LLMs

Lenovo — Raleigh, North Carolina, USA
Intern — 05/2021 - 06/2021

- Collaborated with a team of three other interns to compile and present research on emerging technologies (NFTs, cryptocurrencies) to a Lenovo executive

Publications

- **"On the Necessity of Real-Time Principles in GPU-Driven Autonomous Robots,"** Syed W. Ali, Angelos Angelopoulos, Denver Massey, Sarah Haddix, Alexander Georgiev, Joseph Goh, Rohan Wagle, Prakash Sarathy, James H. Anderson, Ron Alterovitz, *in Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*, March 2025

Activities & Projects

- **A Proportional Share Resource Allocation Algorithm For Real-Time, Time-Shared Systems: A Closer Look**; honors thesis proving time bounds and correctness for the earliest eligible virtual deadline first (EEVDF) scheduling algorithm and analyzing its implementation in the Linux kernel, passed thesis defense with Honors
- **President of The UNC Chapter of the Association of Women in Mathematics**; organized and publicized events such as study nights, alumni/faculty panels, and social events to help foster a sense of community for women in the mathematics department; club is advised by Dr. Caroline Moosmueller
- **Selected for UNC's Undergraduate CyberSecurity Engagement Joint Program with The University of Nagoya in Japan**; exchange program including a trip to Nagoya, Japan to explore the intersection of cybersecurity, national security, and foreign policy through workshops, lectures, and site visits with industry and government leaders
- **Formal Verification of Onion Franking Security Properties Using Tamarin**; semester long project in COMP 590/637 to formally verify certain properties of a metadata-hiding cryptographic protocol that enables anonymous abuse reporting, proposed by researchers at The University of North Carolina in *Onion Franking: Abuse Reports for Mix-Based Private Messaging*, NDSS 2025